

Private markets and inflation pass-through

Status: GOVERNING (2026-08-15; ER-14 filed 2026-08-16). A technical account of how the private market asset classes are modeled across the three layers that model them, and of what happens to a private book when the world's inflation changes.

> **The inflation finding below is filed as ER-14** in > docs/engine-realism-register.md , which is the governing statement of it and > carries the consequences of any fix. This document is the supporting detail — > the layer-by-layer account of the model, the unconsumed-field map, and the > probes that produced the numbers.

The short version. Private markets are modeled in three separate places that agree on the state contract and disagree on almost everything else: the toy-v0 engine's three levered/spread/rate-linked return streams, the generator adapter's sealed factor loadings, and ah/port/ 's cohort cashflow model that sits on top of either tape. The cashflow layer is the detailed one — commitments, calls, a staggered vintage ladder, distributions on a bow, terminal lapse, appraisal-smoothed marks, and a forced-secondary waterfall.

Inflation does not reach any of it. On the shipped player-facing path, moving a world's average inflation from 1% to 12% changes the private equity return by **exactly zero**, changes private credit by +0.02pp/yr and real estate by **-0.12pp/yr**. Every apparent inflation sensitivity in the private programme is a second-order consequence of holding five points of commodities next to it: remove that sleeve and higher inflation makes the private book slightly *smaller*. There is one field in the whole WorldSpec contract that expresses asset-side inflation linkage — structural.infrastructure.inflation_linkage — and it belongs to an asset class the engine does not simulate. Measured numbers and the commands that produce them are in §4 and §7.

1. Three layers, and which one is live

Layer	Module	What it produces	On the player-facing path?
Return process (toy)	ah/core/engine.py	monthly pe / pc / re true + reported returns	yes — toy-v0.6 , the shipped tape
Return process (generated)	ah/port/adapter.py	the same three streams from sealed factor loadings	yes — for generated worlds
Cashflow / institution	ah/port/ + ah/play.py	commitments, calls, distributions, NAV, forced sales	yes — port-v4-ladder

Sealed sleeve mappings (9 sleeves)	ah/port/mapping.py	HF + PM sleeve panels for research	research only
Forward smoothing kernel	ah/port/smoothing.py	fitted per-sleeve reported marks	no — ER-11

The last two matter for reading the evidence documents but do not touch a score.

ah/port/smoothing.py 's own header says so: the product's reported plane is the toy engine's filter, not the fitted kernel, and that divergence is recorded as ER-11.

2. How the return process models each class

2.1 The toy engine (ah/core/engine.py , toy-v0.6)

Three private streams, each a monthly percent return built from the same market state as the public assets. ASSETS is (equity, bonds, hy, commodities, reits, pe, pc, re) — **there is no infrastructure asset**, and reits is dropped on generated worlds (OD-3).

Private equity is levered public equity plus a fixed carry:

```
pe = 1.4 * eq + (illiquidity_premium + entry_multiple_drift) / 12 + 2.0 * e_pe
```

A 1.4× beta to the equity stream, an annual illiquidity premium and a valuation-multiple drift taken from structural.private_equity (defaults 2.0 and 0.0), and an idiosyncratic Student-t innovation. The equity beta is a constant: structural.private_equity.leverage_turns is in the schema and in four presets, and **no engine reads it**, so the leverage a world declares does not change the beta the engine applies.

Private credit is a spread-earning book with a lagged, spread-scaled loss rate:

```
pc = (rate + 4.5)/12 - pc_loss_m - 0.8*d_spread + 0.18*eq + 1.45*e_pc
pc_loss_m = (annual_loss_rate/12) * (0.7 + 0.6 * spread_lagged/400) * crisis_amp
```

Losses key on the HY spread as it stood twelve months earlier and are amplified 1.6× inside crisis months (ER-1's close-out). recovery_rate_pct and spread_over_base_bps are in the schema and unread; the +4.5% credit spread over base is a hardcoded constant.

Real estate is an income stream that reprices on rates:

```
re = 4.5/12 - cap_rate_shift/(100*nm)*2.2 - 4.0*d_rate + 0.35*eq + 1.5*e_re - 1.0*crisis
```

cap_rate_shift_bps is read; income_yield_pct is **not** — the 4.5% income yield is hardcoded, so a world declaring 8% property income gets 4.5%.

Every monthly return is floored at -99% (limited liability, ER-7's close-out).

2.2 The generated path (`ah/port/adaptor.py`)

For generated worlds the same three streams come from the sealed `mappings/sleeve-mappings-v1.0.yaml` PM loadings applied monthly, via `PM_SLEEVE_FOR_ASSET = {pe: pm_buyout, pc: pm_direct_lending, re: pm_re_value_add}`. The regressor set is fixed and is worth reading closely:

```
equity_mkt, smb, hml, mom, d_level, d_slope, d_ig
```

There is no inflation regressor and no commodity regressor. The estimated loadings are also far smaller than the priors they superseded — `pm_buyout` came out at `equity_mkt` 0.348 against a 1.2 prior, and `pm_direct_lending` estimated to *all-zero loadings* with `r2_train_val = -0.0` on 39 quarters, so generated-world private credit is alpha plus noise. `pm_re_value_add` is `equity_mkt` 0.087 with `r2` 0.013. These are recorded honestly in the artifact (each carries its `prior_superseded` block and its own `r2`); the point here is what they imply — on the generated path, two of the three private sleeves are very nearly independent of everything the world does.

2.3 Reported marks

Both paths share `engine._reported_marks`: a Geltner partial adjustment applied at quarter ends only, to the **whole quarter's compounded true return**:

```
rep_q = w * q_true + (1 - w) * rep_{q-1}
```

with `w` per sleeve from `structural.smoothing.weights_on_truth` (defaults 0.35 / 0.30 / 0.35). Unit DC gain, so cumulative reported catches cumulative true. Until `toy-v0.6` this filtered only the closing month and reported PM cumulated about a third of truth — that is ER-10, and it was found by reading a chart, not by a test.

3. How the cashflow layer models a private programme

This is where the modeling actually lives, in `ah/port/` driven by `ah/play.py`. It is the same object model for all three sleeves — the sleeves differ only in their return stream and their target weight, not in their mechanics.

The cohort recursion (`ah/port/cohort.py`) is the model-parameter-register form, quarterly:

```
call_t = RC(age) * f_call * unfunded      (capped at unfunded)
dist_t = Y * (age/L)^B * f_dist * NAV_grown (bounded [0, 1])
NAV_t = NAV*(1+r) + call_t - dist_t        (floored at 0)
```

At `age >= L` (extensions excluded) the fund liquidates in full and the residual unfunded commitment **expires** and is ledgered as `expired_undrawn` — ER-6's close-out, chosen over silently retaining it forever.

Market linkage is tier 1 (`ah/port/cashflow_tier1.py`), frozen in `mappings/cashflow-tier1-v1.0.yaml` before any replay existed:

```
f_dist = clip(exp(-1.541*dd - 1.377*ln(spread_ratio)), 0.3, 1.5)
f_call = clip(1 - 0.1*dd, 0.5, 1.2)
```

`dd` is equity drawdown depth, `spread_ratio` the HY spread over a 400bp reference. Both consume **continuous market states only** — no regime label, no recession dummy, enforced by signature (Delta 3). Distributions dry up hard in stress; calls barely slow. Note what the two arguments are: equity drawdown and credit spread. **Inflation is not an input to either function**, so a stagflation world starves distributions only to the extent that it also crashes equities or widens spreads.

Running the same recursion with `f_call = f_dist = 1` and fees off is tier 0, the constant-G Takahashi–Alexander benchmark — asserted by test, so there are never two models that can disagree about anything except the linkage.

The opening book is a staggered ladder (`play._seed_ladder`): one rung per year of the ten-year contractual life, each warmed forward by the model itself at the rate that reproduces the fixture's own TVPI at its own age, then scaled so the sleeve opens at the intended NAV. Before this (ER-12, closed 2026-08-14) all three sleeves were clones of one age-5.25 cohort and lapsed in the *same quarter*, expiring 17% of the decade's calls at once. A new vintage is committed annually at 18% of the sleeve's target NAV, flexed by the pacing rule toward the policy private weight on **reported** marks.

The institution (`ah/port/engine.py`) runs a quarterly waterfall: cash receives distributions, pays calls, then pays spending at 4.5%/yr on the trailing twelve-quarter average of **reported** total value — so spending holds up in absolute terms exactly when liquid assets are scarcest. If cash goes negative, liquid sleeves sell pro-rata first, then private interests at a 19% secondary haircut, every event logged. Private weight is checked against a (0.15, 0.40) policy band on both the true and the reported basis.

4. Inflation pass-through: the measured answer

4.1 Where inflation exists in the contract

`factor_conditions.inflation` carries `average_pct`, `peak_pct`, `peak_quarter`. On the asset side, the **only** field in the entire WorldSpec that expresses inflation linkage is:

```
structural.infrastructure.inflation_linkage # "Share of revenues contractually inflation-linked", 0..
```

Infrastructure is not in `ASSETS`. The field is reachable by the compiler and the validator's sleeve list and is consumed by no engine. The one place the contract knows how to say "this asset class passes inflation through" is attached to the one private class that is not simulated.

4.2 Where inflation enters the toy engine

Exactly three channels, all keyed on the world's declared **average**, none on the simulated path:

1. **Policy-rate shock size** — `shock = 0.22 * (1 + 0.10 * max(0, infl_avg - 2.0))`. Below 2% inflation this term is identically zero. 2. **Equity–bond correlation sign** — `corr_eb = 0.35 if infl_avg > 3.5 else -0.30`. A step function, and it enters only the bond innovation. 3. **Commodity drift** — `com_drift/12 + max(0, infl_avg - 2.5)/12`.

The simulated inflation *path* (`EnginePaths.inflation`) is generated, carried on the digest, and read only by display surfaces — `cioview.py`, `console.py`, `feed.py`. **No return equation reads it**. Nor does `ah/play.py`, nor any `ah/port/` module.

4.3 Measured: annualized return by declared inflation

Stagflation preset, 200 paths, `base_seed=12345`, one field varied (`factor_conditions.inflation.average_pct`), everything else held:

infl %	equity	bonds	hy	commodities	reits	pe	pc	re
1.0	-0.997	5.314	9.352	11.322	-1.051	-1.772	7.369	1.839
2.0	-0.997	5.314	9.352	11.322	-1.051	-1.772	7.369	1.839
3.4	-0.997	5.288	9.356	12.319	-1.054	-1.772	7.372	1.829
3.6	-0.997	5.304	9.356	12.542	-1.055	-1.772	7.372	1.827
6.5	-0.997	5.236	9.363	15.817	-1.064	-1.772	7.378	1.798
12.0	-0.997	5.059	9.378	22.271	-1.089	-1.772	7.391	1.722

Read the columns:

- **Private equity is bit-identical across a twelvefold change in inflation.** `pe = 1.4*eq + const`, and equity itself carries no inflation term, so the pass-through is not small — it is zero by construction.
- **Private credit gains 0.02pp** over the whole range, via the rate level and the spread-change term. Noise.
- ****Real estate loses 0.12pp**** — the wrong sign for the asset class most often held as an inflation hedge.
- **Commodities is the only asset with material pass-through**, +11pp across the range, and the only one where the mechanism is explicit.

The negative signs on bonds, real estate and REITs are **volatility drag, not repricing** — worth stating precisely, because the intuitive reading (higher inflation raises rates, which marks property down) is not what the code does. The realized rate path is almost unmoved by declared inflation (mean policy rate 6.342 → 6.358 from 1% to 12%; start-to-end change actually *falls*, 1.774 → 1.741). What rises is the rate-shock magnitude, and each asset's compounded return falls in proportion to its own `d_rate` coefficient while its arithmetic mean rises slightly:

asset	<code>d_rate</code> coefficient	Δ compounded	Δ arithmetic	Δ vol
bonds	−6.0	−0.255	+0.055	+4.313
re	−4.0	−0.117	+0.013	+1.367
reits	−2.5	−0.038	+0.008	+0.331
pc	0.0	+0.022	+0.015	+0.006
pe	0.0	0.000	0.000	0.000

So inflation reaches real estate only as extra noise on a duration term, and reaches private equity not at all — zero on the mean, zero on the volatility, zero on the compounded return.

- **Rows 1.0 and 2.0 are identical in every column** — below the 2% anchor the rate-shock term is inert, so a deflationary world and a target world are the same world to this engine.
- `peak_pct` **changes nothing at all**. Holding the average at 6.5 and moving the declared peak from 7.0 to 20.0 leaves every asset's return unchanged to three decimals. `_inflation_path` reads only `average_pct` ; `peak_pct` is consumed solely by the generator's joinery waypoints.

4.4 Measured: the institution

`simulate_play` on the same tape, no player decisions, `seed=12345` , summed over the decade:

infl %	calls	distributions	spending	final NAV	forced-sale quarters
1.0	45.478	46.325	38.444	72.058	28
2.0	45.478	46.325	38.444	72.058	28
6.5	46.487	46.344	39.156	76.012	28
12.0	48.292	46.390	40.417	84.677	28

Final NAV rises 17% from the coldest world to the hottest, which looks like inflation sensitivity. It is not. Move the five points of commodities into equity and re-run:

infl %	private NAV (with commodities)	private NAV (commodities → equity)
1.0	34.982	31.066
6.5	36.016	30.953

12.0	37.917	30.702
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Without the commodity sleeve, the private book gets smaller as inflation rises. The entire positive inflation response of the private programme is an indirect effect of a liquid sleeve sitting next to it in the same portfolio, transmitted through total NAV → reported private weight → the pacing multiplier → commitments → calls. Nothing in the private model itself responds.

Distributions are near-flat across all four worlds (46.33 → 46.39) because the tier-1 linkage reads only equity drawdown and credit spread, and neither moves with declared inflation on this preset.

5. What is declared but not consumed

Fields that exist in `schemas/`, are carried in presets, and reach no engine:

Field	Where it is declared	Status
<code>structural.private_equity.leverage_turns</code>	schema + 4 presets	unread; the 1.4× beta is a constant
<code>structural.private_credit.recovery_rate_pct</code>	schema	unread
<code>structural.private_credit.spread_over_base_bps</code>	schema	unread; +4.5% is hardcoded
<code>structural.real_estate.income_yield_pct</code>	schema	unread; 4.5% is hardcoded
<code>structural.infrastructure.*</code> (both fields)	schema, validator sleeve list	unread; no infra asset exists
<code>structural.smoothing.weights_on_truth.infrastructure</code>	schema	unread; no infra asset exists
<code>factor_conditions.inflation.peak_pct</code>	schema + every preset	unread by <code>toy-v0</code> ; generator joinery only

This is not a defect list — `schemas/` is vendored read-only truth and is allowed to describe more than any one engine implements. It is a map of where a world author's intent currently stops at the contract boundary. A player or an author who sets `income_yield_pct: 8.0` and `leverage_turns: 7.0` will see no change in any number.

6. What this means, and what it does not

It does not invalidate anything. Every mechanism above is faithful to its plan, and several of them (ER-1, ER-6, ER-7, ER-10, ER-12) are close-outs of defects that were found and fixed. The cashflow layer in particular is detailed, honest about its own choices, and the part of the system most worth trusting.

It does mean the platform cannot currently answer an inflation question about private markets. The stagflation preset is one of the four shipped worlds and the one the validation battery runs on. A player allocating in it, or a reader of a stagflation chronicle, will see a private book that is — measurably — indifferent to the defining feature of that world. Real allocators hold infrastructure and real estate *specifically* for inflation linkage; here that linkage is absent in the returns, absent in the cashflows, absent in the distribution response, and inverted in sign for property.

Spending is nominal too. The institution spends 4.5% of nominal reported value with no CPI indexation, so a stagflation world silently erodes real spending power rather than forcing the liquidity squeeze a real inflation-indexed payout would. Note that `ah/port/twin.py` — the *pension* twin, off the play path — does index: `inflation_linked_share` defaults to 0.7 and `inflation_shock()` moves liability PV with an `inflation_hedge_ratio` offset. The machinery exists; the played institution does not use it.

Filed as ER-14 (owner decision, 2026-08-16). The finding is register-shaped — faithful to plan, not faithful to an allocator's expectations — and `docs/engine-realism-register.md` **ER-14 is now the governing statement of it**, including the consequences of any fix: `TOY_ENGINE_VERSION` and both `PLAY_ALPHA_VERSION` stamps bump, every `RunRecord` digest is invalidated, both committed bundles rebuild, the battery re-runs, and admitting inflation to tier 1's linkage would need an amendment to a sealed artifact plus a decision about the Delta 3 no-regime-label rule. Status is `open`, no fix scheduled. This document remains the supporting detail: the layer-by-layer account, the unconsumed-field map, and the probes in §7.

7. Reproducing the numbers

Every figure above comes from three short probes against committed presets and artifacts; no fixtures, no network. From the repo root:

```
# §4.3 – return by declared inflation
import copy, json, numpy as np
from ah.core.loader import load_worldspec
from ah.core.numericworld import project_numeric
from ah.core.engine import run_ensemble, ASSETS

base = json.load(open("src/ah/presets/stagflation.json", encoding="utf-8"))
for infl in (1.0, 2.0, 6.5, 12.0):
    d = copy.deepcopy(base)
    d["factor_conditions"]["inflation"]["average_pct"] = infl
    ens = run_ensemble(project_numeric(load_worldspec(d)), 200, base_seed=12345)
```



```

for a in ASSETS:
    r = ens.returns[a] / 100.0
    print(infl, a, (np.prod(1 + r, axis=1).mean() ** (12 / r.shape[1]) - 1) * 100)

# §4.4 – the institution, and the commodity attribution
from ah.core.engine import run_path
from ah.play import simulate_play, PRIVATE_ASSETS, START_TARGETS

targets = dict(START_TARGETS)          # drop the commodity sleeve
targets["equity"] += targets["commodities"]; targets["commodities"] = 0.0
for infl in (1.0, 6.5, 12.0):
    d = copy.deepcopy(base)
    d["factor_conditions"]["inflation"]["average_pct"] = infl
    tape = run_path(project_numeric(load_worldspec(d)), 12345)
    for label, tg in (("with", None), ("without", targets)):
        res = simulate_play(tape, {}, start_targets=tg)
        last = res.quarters[-1]
        print(infl, label, sum(last.private_true[a] for a in PRIVATE_ASSETS))

```

Reading, rather than running: `ah/core/engine.py:405-448` (the return block), `:288-296` (`_inflation_path`), `:451-485` (`_reported_marks`); `ah/port/cohort.py:163-227` (the recursion); `ah/port/cashflow_tier1.py:48-66` (the linkage); `ah/port/engine.py:86-183` (the waterfall); `ah/play.py:594-677` (the quarterly loop); `mappings/cashflow-tier1-v1.0.yaml` and `mappings/sleeve-mappings-v1.0.yaml` (the frozen parameters).

Related

- `docs/engine-realism-register.md` — **ER-14** is this document's finding, filed. ER-1, ER-3, ER-6, ER-7, ER-10, ER-11 and ER-12 all touch the same surface; ER-3's "what is still open" note is the closest prior statement of the boundary described here.
- `Instructions/DN1.1-multiyear-generator-design-note.md`, DN-5 §3.3 — the PM growth-loading priors that `sleeve-mappings-v1.0.yaml` superseded.
- `G1-EVIDENCE.md` — the Step 3 completion gate, an honest FAIL, with tier 1 beaten by tier 0 on the 2022 episode.